

Hands on 3

Object to Relational Mapping in Hibernate using Annotations (Employee Class)

In Hibernate with **annotation-based configuration**, object-relational mapping is done directly in the Java class using annotations from the jakarta.persistence (or older javax.persistence) package.

◆ Example: Mapping Employee Class to Database Table

```
import jakarta.persistence.*;
```

```
@Entity
```

```
@Table(name = "EMPLOYEE")
```

```
public class Employee {
```

```
    @Id
```

```
    @GeneratedValue(strategy = GenerationType.IDENTITY)
```

```
    @Column(name = "id")
```

```
    private int id;
```

```
    @Column(name = "first_name")
```

```
    private String firstName;
```

```
    @Column(name = "last_name")
```

```
    private String lastName;
```

```
    @Column(name = "salary")
```

```
    private int salary;
```

```
    // Getters and Setters
```

}

◆ Explanation of Annotations:

Annotation	Purpose
@Entity	Marks the class as a Hibernate entity (table-mapped class).
@Table(name="EMPLOYEE")	Maps the class to a specific table in the database (here, EMPLOYEE).
@Id	Marks the field as the primary key .
@GeneratedValue	Automatically generates value for the primary key. Can specify strategy like AUTO, IDENTITY, SEQUENCE.
@Column	Maps the Java field to the corresponding column name in the database.

◆ Hibernate Configuration File (hibernate.cfg.xml)

This file provides the Hibernate engine with necessary settings to connect to your database and locate your entity classes.

◆ Example:

```
<hibernate-configuration>

  <session-factory>

    <!-- JDBC Database connection settings -->

    <property
name="hibernate.connection.driver_class">com.mysql.cj.jdbc.Driver</property>

    <property
name="hibernate.connection.url">jdbc:mysql://localhost:3306/your_db</property>

    <property name="hibernate.connection.username">root</property>

    <property name="hibernate.connection.password">your_password</property>

    <!-- JDBC dialect -->

    <property name="hibernate.dialect">org.hibernate.dialect.MySQLDialect</property>
```

```
<!-- Hibernate settings -->

<property name="hibernate.show_sql">true</property>

<property name="hibernate.hbm2ddl.auto">update</property>


<!-- Register annotated class -->

<mapping class="com.yourpackage.Employee"/>

</session-factory>

</hibernate-configuration>
```

◆ Explanation of Configuration Elements:

Element / Property	Purpose
hibernate.connection.driver_class	The JDBC driver for your database (e.g., MySQL).
hibernate.connection.url	JDBC URL for connecting to your DB (includes host, port, and DB name).
hibernate.connection.username	Your DB username (e.g., root).
hibernate.connection.password	Password for the DB user.
hibernate.dialect	Tells Hibernate what type of DB you're using (e.g., MySQL).
hibernate.show_sql	Logs SQL queries to the console (for debugging).
hibernate.hbm2ddl.auto	Auto-create or update tables: values can be update, create, etc.
<mapping class="...">	Registers your annotated entity class with Hibernate.